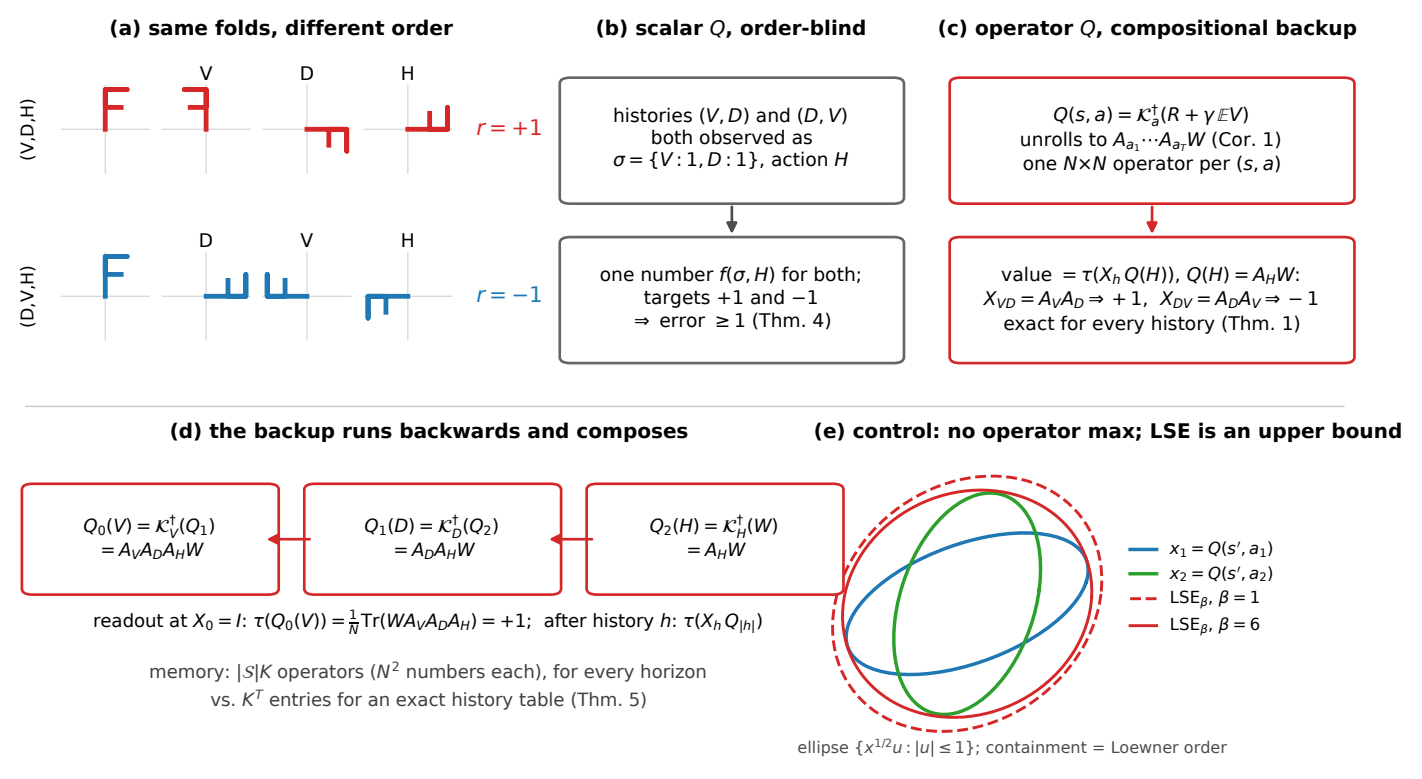
(a) same folds, different order
(V,D,H)
V
D
H
r = +1
(D,V,H)
D
V
H
r = −1
(b) scalar Q, order-blind
histories (V, D) and (D, V) both observed as σ = {V : 1, D : 1}, action H
one number f(σ, H) for both; targets +1 and −1
⇒ error ≥ 1 (Thm. 4)
(c) operator Q, compositional backup
Q(s, a) = K_a^†(R + γ E V) unrolls to A_{a_1} ··· A_{a_τ} W (Cor. 1)
one N×N operator per (s, a)
value = τ(X_h Q(H)), Q(H) = A_H W:
X_VD = A_V A_D ⇒ +1, X_DV = A_D A_V ⇒ −1
exact for every history (Thm. 1)
(d) the backup runs backwards and composes
Q_2(H) = K_H^†(W) = A_H W
Q_1(D) = K_D^†(Q_2) = A_D A_H W
Q_0(V) = K_V^†(Q_1) = A_V A_D A_H W
readout at X_0 = I: τ(Q_0(V)) = (1/N) Tr(W A_V A_D A_H) = +1; after history h: τ(X_h Q_{|h|})
memory: |S|K operators (N^2 numbers each), for every horizon
vs. K^T entries for an exact history table (Thm. 5)
(e) control: no operator max; LSE is an upper bound
x_1 = Q(s′, a_1)
x_2 = Q(s′, a_2)
LSE_β, β = 1
LSE_β, β = 6
ellipse {x^{1/2} u : |u| ≤ 1}; containment = Loewner order